Ahmedur Rahman Shovon

Postdoctoral Appointee, Argonne National Laboratory

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Google scholar

Research Interests

High-Performance Computing	Runtime Systems, Parallel Programming Models, Agentic Workflows
Data Analytics	Declarative Analytics, Topological Data Analysis, Data Visualization
Performance Engineering	Performance Monitoring, Energy Efficiency, Cloud and Edge Systems

Education

PhD in Computer Science **08/2025**
University of Illinois Chicago *Chicago, IL*
Dissertation: Declarative Analytics on Heterogeneous HPC Systems
Advisor: Dr. Sidharth Kumar

MS in Information Technology **11/2018**
Jahangirnagar University *Dhaka, Bangladesh*
Thesis: A RESTful E-Governance Application Framework for People Identity Verification in Cloud
Advisor: Dr. Md Whaiduzzaman

BS in Information Technology **04/2016**
Jahangirnagar University *Dhaka, Bangladesh*
Project: EasyC: A Platform for Learning C Programming
Advisor: Dr. Risala Tasin Khan

Experience

Postdoctoral Appointee **08/2025 - p.**
Mathematics and Computer Science Division, Argonne National Laboratory *Lemont, IL*

Graduate Research and Teaching Assistant **08/2023 - 08/2025**
Department of Computer Science, University of Illinois Chicago *Chicago, IL*

Graduate Student Intern (WJ Cody Associate) **05/2023 - 08/2023**
Mathematics and Computer Science Division, Argonne National Laboratory *Lemont, IL*

Graduate Research Assistant (Blazer Fellow) **08/2021 - 08/2023**
Department of Computer Science, University of Alabama at Birmingham *Birmingham, AL*

Assistant Programmer **06/2019 - 08/2021**
ICT Division, Government of Bangladesh *Dhaka, Bangladesh*

Software Engineer **12/2017 - 06/2019**
Cefalo Bangladesh Ltd. *Dhaka, Bangladesh*

Software Engineer **10/2016 - 11/2017**
Codalo *Dhaka, Bangladesh*

Teaching Experience

Graduate Teaching Assistant **Spring 2025**
Department of Computer Science, University of Illinois Chicago *Chicago, IL*
Course: CS 455 Introduction to High-Performance Computing (class size: ∼60)

Graduate Teaching Assistant **Fall 2024**
Department of Computer Science, University of Illinois Chicago *Chicago, IL*
Course: CS 480 Database Systems (class size: ∼80)

Graduate Teaching Assistant **Fall 2016**
Institute of Information Technology, Jahangirnagar University *Dhaka, Bangladesh*
Course: IT 3107 Operating Systems (class size: ∼60)

Publications

Journal Articles (J)

[**J15**] Ahmedur Rahman Shovon, Sidharth Kumar, Gopikrishna Deshpande. **Topology Assisted Clustering of Temporal fMRI Brain Networks With Use-Case in Mitigating Non-Neural Multi-Site Variability**. *IEEE Access, 2025.* DOI: 10.1109/ACCESS.2025.3616256.

[**J14**] Sidharth Kumar, Ahmedur Rahman Shovon, Gopikrishna Deshpande, **The robustness of persistent homology of brain networks to data acquisition-related non-neural variability in resting state fMRI**. *Human Brain Mapping, 2023.* DOI: 10.1002/hbm.26403.

[**J13**] Darshan Shimoga Chandrashekar, Santhosh Kumar Karthikeyan, Praveen Kumar Korla, Henalben Patel, Ahmedur Rahman Shovon, Mohammad Athar, George J Netto, Zhaohui S Qin, Sidharth Kumar, Upender Manne, Chad J Crieghton, Sooryanarayana Varambally. **UALCAN: An update to the integrated cancer data analysis platform**. *Neoplasia, 2022.* DOI: 10.1016/j.neo.2022.01.001.

[**J12**] Md Whaiduzzaman, Md. Razon Hossain, Ahmedur Rahman Shovon, Shanto Roy, Aron Laszka, Rajkumar Buyya, Alistair Barros. **A Privacy-preserving Mobile and Fog Computing Framework to Trace and Prevent COVID-19 Community Transmission**. *IEEE Journal of Biomedical and Health Informatics (J-BHI), 2020.* DOI: 10.1109/JBHI.2020.3026060.

Book Chapters (B)

[**B11**] Ahmedur Rahman Shovon, Shanto Roy, Tanusree Sharma, Md. Whaiduzzaman. **A RESTful E-Governance Application Framework for People Identity Verification in Cloud**. *Luo M., Zhang LJ. (eds) Cloud Computing – CLOUD 2018. Lecture Notes in Computer Science, vol 10967. Springer, Cham.* DOI: 10.1007/978-3-319-94295-7_19.

Conference Papers (C)

[**C10**] Ahmedur Rahman Shovon, Yihao Sun, Thomas Gilray, Kristopher Micinski, Sidharth Kumar. **Multi-Node Multi-GPU Datalog**. *ACM International Conference on Supercomputing.*
ICS 2025, DOI: 10.1145/3721145.3730431.

[**C9**] Yihao Sun, Ahmedur Rahman Shovon, Thomas Gilray, Kristopher Micinski, Sidharth Kumar. **Optimizing Datalog for the GPU**, *ACM International Conference on Architectural Support for Programming Languages and Operating Systems.*
ASPLOS 2025, DOI: 10.1145/3669940.3707274.

[**C8**] Andres Sewell, Ke Fan, Ahmedur Rahman Shovon, Landon Dyken, Sidharth Kumar, Steve Petruzza. **Bruck Algorithm Performance Analysis for Multi-GPU All-to-All Communication**. *International Conference on High Performance Computing in Asia-Pacific Region.*
HPCAsia 2024. DOI: 10.1145/3635035.3635047.

[**C7**] Ahmedur Rahman Shovon, Thomas Gilray, Kristopher Micinski, Sidharth Kumar. **Towards Iterative Relational Algebra on the GPU**. *USENIX Annual Technical Conference.*
USENIX ATC 2023, ISBN: 978-1-939133-35-9.

[**C6**] Md Whaiduzzaman, Alistair Barros, Ahmedur Rahman Shovon, Md Razon Hossain, Colin Fidge. **A Resilient Fog-IoT Framework for Seamless Microservice Execution**. *IEEE International Conference on Services Computing.*
SCC 2021 DOI: 10.1109/SCC53864.2021.00034.

[**C5**] Md Shahriare Satu, Md Khalilur Rahman, Maksud Alam Rony, Ahmedur Rahman Shovon, Md Jane Alam Adnan, Koushik Chandra Howlader, M Shamim Kaiser. **COVID-19: Update, Forecast and Assistant-An Interactive Web Portal to Provide Real-Time Information and Forecast COVID-19 Cases in Bangladesh**. *International Conference on Information and Communication Technology for Sustainable Development.*
ICICT4SD 2021, DOI: 10.1109/ICICT4SD50815.2021.9396786.

[**C4**] Ahmedur Rahman Shovon, Shanto Roy, Arnab Kumar Shil, Tanjila Atik. **GDPR Compliance: Implementation Use Cases for User Data Privacy in News Media Industry**. *International Conference on Advances in Science, Engineering and Robotics Technology.*
ICASERT 2019, DOI: 10.1109/ICASERT.2019.8934660.

[**C3**] Shanto Roy, Ahmedur Rahman Shovon, Md. Whaiduzzaman. **Combined approach of Tokenization and Mining to secure and optimize Big Data in Cloud Storage**. *IEEE R10 Humanitarian Technology Conference.*
R10-HTC 2017, DOI: 10.1109/R10-HTC.2017.8288912.

Workshop Papers (W)

[**W2**] Kashyap Balakavi, Rushitha Janga, Ahmedur Rahman Shovon, Don Dempsey, Elliot Lefkowitz, Sidarth Kumar, **Scalable, interactive and hierarchical visualization of virus taxonomic data**. *IEEE Workshop on Visual Analytics in Healthcare.*
VAHC 2023, held with IEEE VIS 2023, DOI: 10.1109/VAHC60858.2023.00012.

[**W1**] Ahmedur Rahman Shovon, Landon Richard Dyken, Oded Green, Thomas Gilray, Sidharth Kumar. **Accelerating Datalog applications with cuDF**. *Workshop on Irregular Applications: Architectures and Algorithms.*
IA3 2022, held with SC22, DOI: 10.1109/IA356718.2022.00012.

Awards

USENIX ATC Student Grant 2023
Awarded by the USENIX Association.

Blazer Graduate Research Fellowship 2021-2022
Awarded by the University of Alabama at Birmingham.

National Science and Technology (NST) Fellowship 2016-2017
Awarded by the Ministry of Science and Technology, Bangladesh.

University Merit Scholarships 2012-2016
Awarded by the Jahangirnagar University, Bangladesh.

Achievements

3rd @ Computational Research Symposium Poster Presentation 2025 04/2025
Presented posted on our multi-node multi-GPU Datalog engine (MNMGDatalog).

Winner @ oneAPI Hackathon: CUDA to SYCL Migration 10/2023
Ported CUDA project to SYCL with automatic and manual interventions.

1st @ Hackerrank Python and Security domain 07/2016
Solved all Python and Security domain related challenges at Hackerrank.

7th @ IEEEXtreme Programming Competition 9.0 10/2015
Achieved 7th position out of 2,477 teams, Team: *JUIITCoders*.

Technical Skills

Languages	Python, C++, JavaScript, Java, SQL, Bash, PHP
Parallel Computing	CUDA, MPI, SYCL, HIP, OpenMP, Thrust
Data Analytics	Pandas, NumPy, Scikit-learn, CuDF, Gudhi, Matplotlib, D3.js
Web and Cloud	Flask, Django, React, jQuery, CodeIgniter, Redis, AWS
CI/CD	Docker, GitHub Actions, Travis CI, Jenkins, Boto3, Kubernetes

Professional Service

Journal Reviewer

- IEEE Access
- Sustainable Cities and Society (Elsevier)

Program Committee / Reviewer

- IEEE Cloud Summit (AI/ML for Cloud and IoT track) 2026
- International Conference on Machine Intelligence and Emerging Technologies 2026
- ACM/IEEE Supercomputing Conference (SC) (Subreviewer) 2023, 2025

Proposal Reviewer

- DOE WDTS Visiting Faculty Program 2026

Competition and Expo Judge

- DePaul University DemonHacks 2026
- University of Illinois Chicago SparkHacks 2026
- University of Illinois Chicago Undergraduate Research Forum 2024
- Research Expo, University of Alabama at Birmingham 2021–2023

Open-Source Contributions

- Top 3% contributor on Stack Overflow *Username: arshovon*
- Python Package Index (PyPI) Contributor *Username: arsho*

Outreach and Leadership

- Volunteer, The World Games, Birmingham, AL 2022
- Webmaster, IEEE Student Branch, Jahangirnagar University, Bangladesh 2015–2016